# ⚡ Lab 6: Configure High Availability

## AWS Technical Essentials

Load Balancing, Auto Scaling & Self-Healing

# Resource Names for This Lab

Replace <ID> with your assigned student ID (for example user7).

| Resource | Name |
| --- | --- |
| Application Load Balancer | employee-directory-alb-<ID> |
| ALB security group | employee-directory-alb-sg-<ID> |
| Target group | employee-directory-tg-<ID> |
| Launch template | employee-directory-lt-<ID> |
| Auto Scaling group | employee-directory-asg-<ID> |
| Instances it launches (Name tag) | employee-directory-asg-app-<ID> |
| *Reused from earlier labs* | employee-directory-<ID>-vpc, employee-directory-vpc-sg-<ID>, employee-dir-key-<ID> |

⚠️ **32-character limit:** AWS caps load balancer and target group

# 📋 Lab Overview

## 🎯 Objectives

- Create an Application Load Balancer
- Create a Launch Template
- Configure an Auto Scaling Group
- Test self-healing
- View CloudWatch metrics

## 📝 Lab Details

- **Duration:** ~50 minutes
- **Capstone:** Lab 6 of 6

# ✅ Prerequisites

> **Before starting:** Verify these resources from previous labs exist.

- **VPC:** employee-directory-<ID>-vpc with 2 public subnets in different AZs
- **Security Group:** employee-directory-vpc-sg-<ID> *(the one in your VPC, created in Lab 3)*
- **Key Pair:** employee-dir-key-<ID>

> 💡 **Tip:** Terminate any running EC2 instances from earlier labs to stay within limits.

# 🌐 Verify Your Subnets

You enabled this in **Lab 3**. Confirm it before building anything — the Auto Scaling group inherits it, and fixing it later means replacing every instance.

1. **VPC** → **Subnets**, filter by `<ID>`
2. Check the **Auto-assign public IPv4 address** column for both of your public subnets

> ✅ **Expected:** both show **Yes**. If either shows **No**: **Actions** → **Edit subnet settings** → tick the box → **Save**.

# 🔧 Create the Application Load Balancer

1. Navigate to **EC2** → **Load Balancers** (left nav)
2. Click **Create load balancer** → **Application Load Balancer** → **Create**

| Setting | Value |
|---|---|
| Name | `employee-directory-alb-<ID>` |
| Scheme | **Internet-facing** |
| VPC | **employee-directory-<ID>-vpc** |
| Availability Zones | Check **both** public subnets |

Create a **new security group** for the ALB:

- Name: `employee-directory-alb-sg-<ID>`
- Inbound rule: **HTTP / 80** from `0.0.0.0/0`

# 🔧 Create Target Group and Finish ALB

Under **Listeners and routing**, click **Create target group**:

| Setting | Value |
|---|---|
| Target type | **Instances** |
| Name | `employee-directory-tg-<ID>` |
| Protocol / Port | HTTP / 80 |
| VPC | employee-directory-<ID>-vpc |
| Health check path | `/` |

1. Click **Next**, skip registering targets, click **Create target group**
2. Return to the ALB tab, **refresh**, select **employee-directory-tg-<ID>**
3. Click **Create load balancer**

> ✅ **Expected:** ALB created. Status will show **Provisioning** for a few minutes.

# Create a Launch Template

1. Navigate to **EC2 → Launch Templates**
2. Click **Create launch template**

| Setting | Value |
|---|---|
| Name | `employee-directory-lt-<ID>` |
| AMI | **Amazon Linux 2023** |
| Instance type | **t2.micro** |
| Key pair | **employee-dir-key-<ID>** |
| Security group | **employee-directory-vpc-sg-<ID>** — check the VPC column |
| Resource tag (Name) | `employee-directory-asg-app-<ID>` |

> ⚠️ **Tag the instances:** in **Resource tags**, add **Key** Name / **Value** `employee-directory-asg-app-<ID>` and apply it to **Instances and Volumes**. Without it, every student's Auto Scaling instances appear nameless in a shared account and you cannot tell yours apart.

# Launch Template — User Data

Paste into the **User data** field:

```
#!/bin/bash
dnf install -y httpd
systemctl enable --now httpd
TOKEN=$(curl -sX PUT http://169.254.169.254/latest/api/token -H "X-aws-ec2-metadata-token-ttl-seconds: 21600")
IID=$(curl -s -H "X-aws-ec2-metadata-token: $TOKEN" http://169.254.169.254/latest/meta-data/instance-id)
AZ=$(curl -s -H "X-aws-ec2-metadata-token: $TOKEN" http://169.254.169.254/latest/meta-data/placement/availability-zone)
cat > /var/www/html/index.html <<EOF
<!DOCTYPE html>
<html><head><title>Employee Directory</title>
<style>body{font-family:Arial,sans-serif;max-width:800px;margin:50px auto;padding:20px}
h1{color:#1a365d}table{border-collapse:collapse;width:100%}th,td{border:1px solid #ddd;padding:12px;text-align:left}
th{background:#2b6cb0;color:white}tr:nth-child(even){background:#f7fafc}</style></head>
<body><h1>Employee Directory</h1>
<p>Instance: ${IID:-unknown} | AZ: ${AZ:-unknown}</p>
<table><tr><th>Name</th><th>Department</th><th>Location</th></tr>
<tr><td>John Smith</td><td>Engineering</td><td>Seattle</td></tr>
<tr><td>Jane Doe</td><td>Marketing</td><td>New York</td></tr>
<tr><td>Bob Johnson</td><td>Sales</td><td>Austin</td></tr></table>
</body></html>
EOF
```

1. Click **Create launch template**

# 📊 Create Auto Scaling Group

1. Navigate to **EC2 → Auto Scaling Groups**
2. Click **Create Auto Scaling group**
3. Name: `employee-directory-asg-<ID>`
4. Launch template: **employee-directory-lt-<ID>**
5. Click **Next**
6. VPC: **employee-directory-<ID>-vpc**
7. Subnets: Select **both** public subnets
8. Click **Next**

# 📊 ASG — Attach to ALB

1. Select **Attach to an existing load balancer**
2. Choose **Choose from your load balancer target groups**
3. Select **employee-directory-tg-<ID>**
4. Health check type: Check **Turn on Elastic Load Balancing health checks**
5. Health check grace period: **300** seconds — leave the default, it gives Apache time to install before the first check
6. Click **Next**

| Setting | Value |
|---|---|
| Desired capacity | **2** |
| Min | **2** |
| Max | **4** |

6. Click **Next** through remaining pages, then **Create Auto Scaling group**

# 🌐 Verify — Test the ALB

1. Go to **EC2** → **Instances** — confirm 2 new instances are **Running**
2. Go to **Load Balancers** → select your ALB
3. Copy the **DNS name**
4. Open in a browser: `http://<ALB_DNS_NAME>`
5. Refresh the page several times

> ✅ **Expected:** The Employee Directory page loads. Refreshing shows different **Instance IDs** and **AZs** — the ALB is distributing traffic.

# 🔨 If the Targets Never Go Healthy

Instances launching and terminating on a loop, or the ALB returning **503**, all trace back to the same thing: the target group never sees a **200** on `/`. Work down this list.

| Check | Fix |
|---|---|
| Target group shows *unhealthy* or *draining* | EC2 → Target Groups → Targets tab reports the reason in plain text — start there |
| Launch template security group | Must be `employee-directory-vpc-sg-<ID>`, the one in *your VPC*. The Lab 2 group has a similar name but lives in the default VPC, and the launch fails outright |
| Instances have no public IPv4 | Both public subnets need auto-assign enabled (Lab 3; re-checked on slide 5). Without it there is no route to the repos, so Apache never installs |
| Instance is running but serves 403 | `index.html` was never written. Check `/var/log/cloud-init-output.log` |

# 🔨 Test Self-Healing

1. Go to **EC2 → Instances**
2. Select one of the ASG instances
3. Click **Instance state → Terminate instance**
4. Watch the **Auto Scaling Groups** activity tab

> ✅ **Expected:** Within 1-2 minutes, Auto Scaling launches a **replacement instance** to maintain the desired count of 2. The ALB continues serving traffic through the remaining healthy instance.

# 📊 View CloudWatch Metrics

1. Go to **EC2** → select an instance → **Monitoring** tab
2. Review: CPU utilization, network in/out, status checks
3. Go to **Load Balancers** → select ALB → **Monitoring** tab
4. Review: request count, healthy host count, response time

> 💡 **Key metrics:** The ALB **Healthy Host Count** dropped to 1 when you terminated an instance and returned to 2 when the replacement launched.

# 🗑️ Clean Up — Step 1

> ⚠️ **Delete in this order** to avoid dependency errors:

1. **Auto Scaling Group:** EC2 → Auto Scaling Groups → select → Delete (set desired/min to 0, then delete)
2. **Wait** for all ASG instances to terminate
3. **Load Balancer:** EC2 → Load Balancers → select → Delete
4. **Target Group:** EC2 → Target Groups → select → Delete

# 🗑️ Clean Up — Step 2

5. **Launch Template:** EC2 → Launch Templates → select → Delete
6. **Security Groups:** Delete `employee-directory-alb-sg-<ID>` and `employee-directory-vpc-sg-<ID>`
7. **VPC:** VPC → Your VPCs → select employee-directory-<ID>-vpc → Delete (this removes subnets, route tables, and IGW)
8. **Key Pair:** EC2 → Key Pairs → delete `employee-dir-key-<ID>`
9. **Any remaining EC2 instances** from earlier labs → Terminate

> ⚠️ If VPC deletion fails, check for remaining ENIs or instances. All dependencies must be removed first.

# 🎉 Lab 6 Complete!

## High Availability Architecture Deployed

# 🎯 Key Takeaways

## 🛠️ What You Built

- Application Load Balancer across 2 AZs
- Launch Template with bootstrap script
- Auto Scaling Group with self-healing
- CloudWatch monitoring

## 📚 Key Concepts

- **ALB** distributes traffic across AZs
- **ASG** maintains desired instance count
- **Self-healing** replaces failed instances
- **CloudWatch** provides monitoring

# 🏆 All Labs Complete!

## AWS Technical Essentials

You have built and deployed across IAM, EC2, VPC, S3, DynamoDB, ALB, and Auto Scaling



# Questions?

Final questions before we wrap up the course